Unit 4 Notes: Solving Quadratics by Factoring

Read each section before the matching homework problems.

§1 Standard form

A quadratic equation can be written in standard form $ax^2 + bx + c = 0$, where a is not 0. Before factoring, move every term to one side so the other side is exactly 0. For example, $x^2 + 4x = 12$ becomes $x^2 + 4x - 12 = 0$.

§2 Zero product property

If the product of two factors is 0, then at least one of the factors must be 0. So if (x + 4)(x - 1) = 0, then x + 4 = 0 or x - 1 = 0, which gives x = -4 or x = 1. Set EACH factor equal to zero. Most quadratics have two solutions (roots), and stopping after one is the most common mistake.

§3 Factoring $x^2 + bx + c$

To factor $x^2 + bx + c$, look for two numbers that multiply to c and add to b. Example: $x^2 + 7x + 10$. The numbers 2 and 5 multiply to 10 and add to 7, so $x^2 + 7x + 10 = (x + 2)(x + 5)$. Watch the signs: if c is negative, one number is positive and one is negative.

§4 Common factor first

Always check for a greatest common factor before anything else. Example: $3x^2 + 12x = 0$ factors as $3x(x + 4) = 0$. Never divide both sides by x — you would lose the solution $x = 0$.

§5 Difference of squares

An expression of the form $a^2 - b^2$ factors as $(a - b)(a + b)$. Example: $x^2 - 49 = (x - 7)(x + 7)$. This gives two roots with the same size and opposite signs.

§6 Checking and word problems

Check every root by substituting it into the ORIGINAL equation. In word problems, write down what the variable means, build the equation from the situation, and then decide whether each root makes sense. A negative length or time is not a valid answer, so state why you reject it.